

Thinking like a CHEMIST: Combined Heterogeneous Embedding Model Integrating Structure and Tokens

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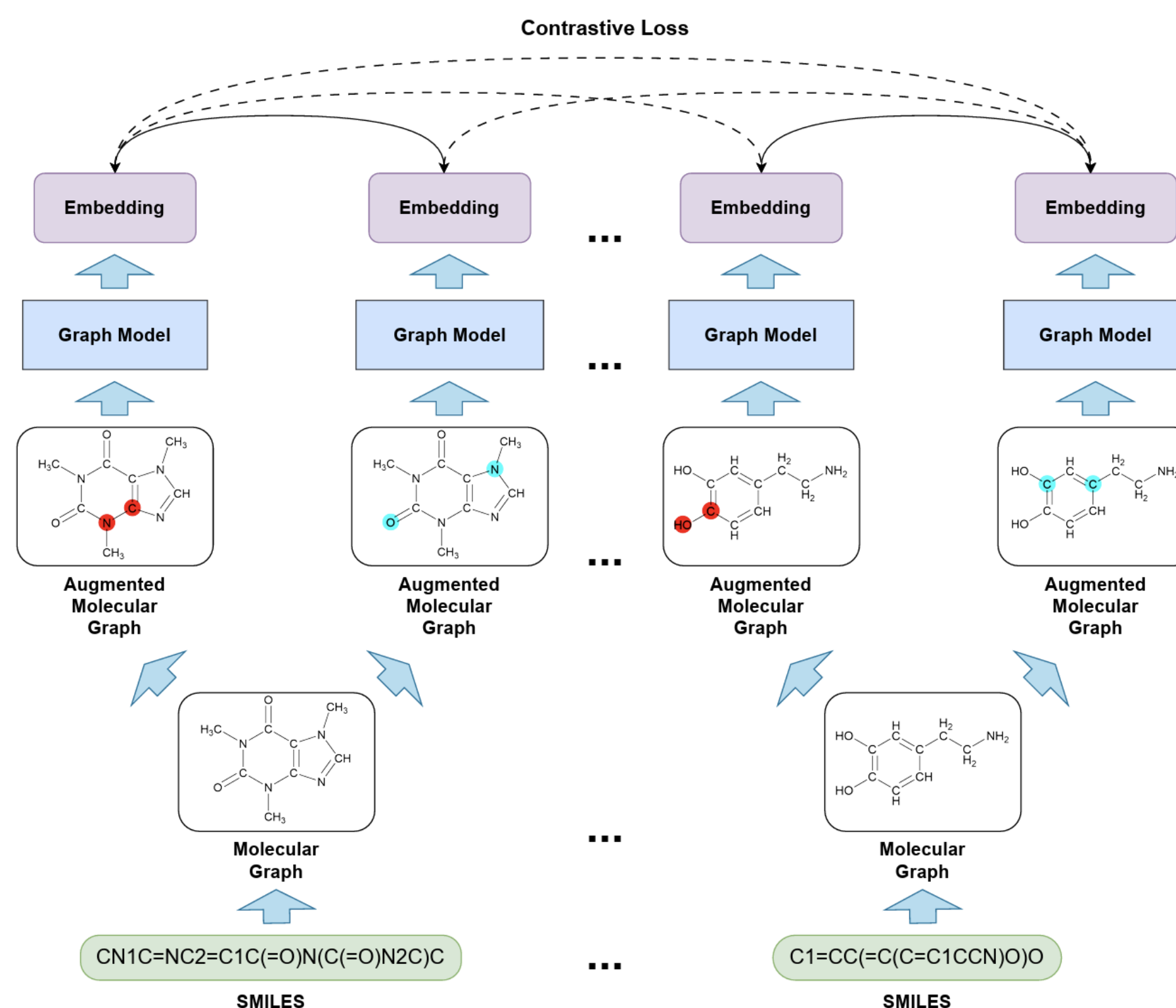
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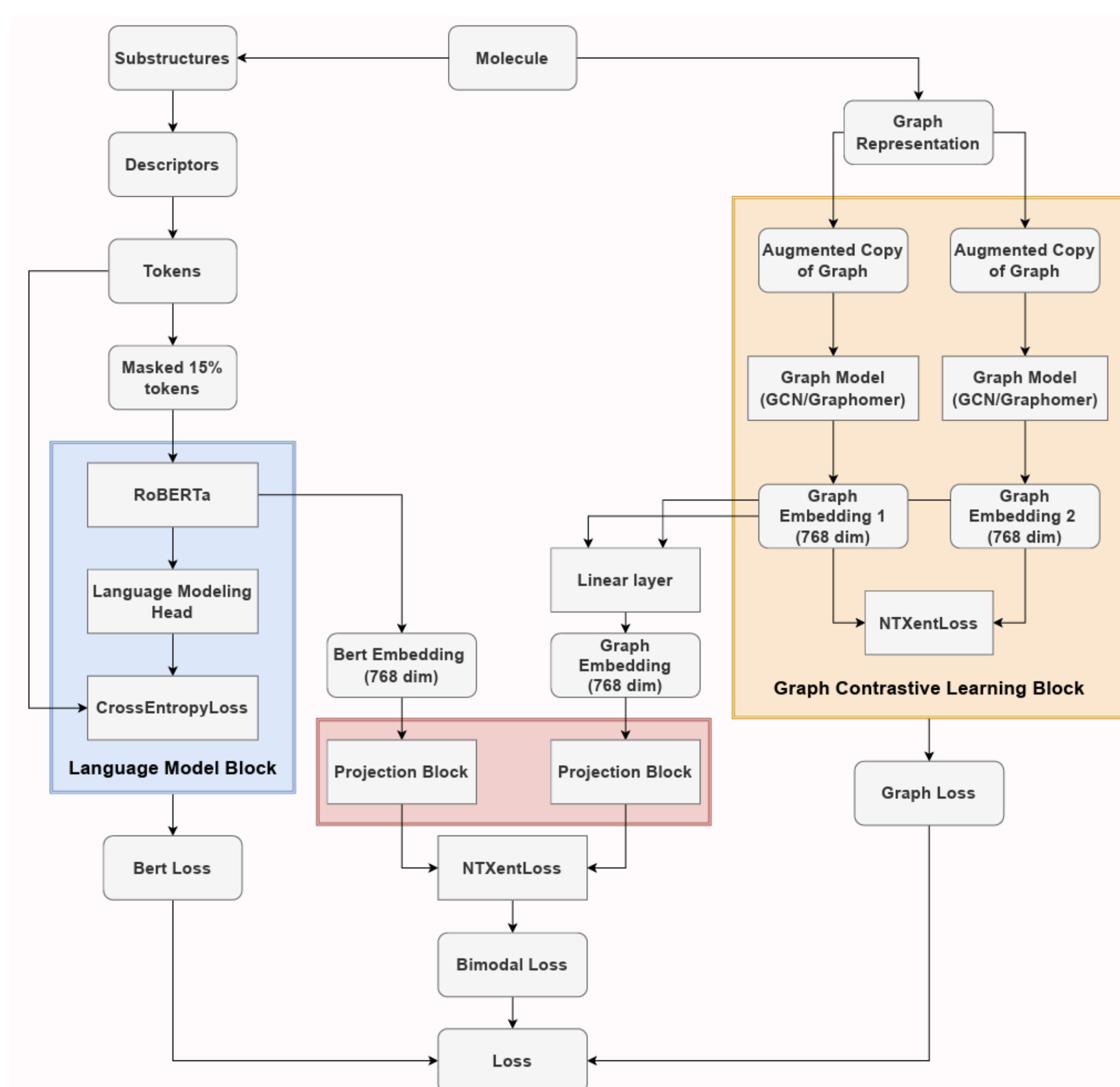
Motivation & Main Contribution

- ❑ Molecular ML struggles to capture both chemical structure and textual (SMILES) semantics
- ❑ Graph and language models each miss complementary chemical cues
- ❑ Need representations closer to how **chemists** reason, via substructures and properties
- We propose **CHEMIST**, combining graph and language embeddings of molecular fragments
- Use **descriptor-based fragment tokens** and **contrastive alignment** for unified molecular understanding

Argumentation Process



Proposed Method



Results

ROC-AUC metric for the classification tasks

Models	Datasets						
	BBBP	Tox21 (NR-AR)	ClinTox (FDA APPROVED)	ClinTox (CT TOX)	BACE	MUV	HIV
MolCLR(GCN) [53]	0.723±0.025	0.704±0.002	0.668±0.035	0.694±0.032	0.711±0.090	0.676±0.019	0.787±0.005
MolCLR(GIN) [53]	0.742±0.020	0.740±0.003	0.872±0.031	0.775±0.037	0.814±0.07	0.796±0.017	0.761±0.006
ChemBERTa [6]	0.647±0.053	0.753±0.009	-	0.736±0.015	0.721±0.022	0.667±0.015	0.625±0.012
Uni-Mol [62]	0.729±0.006	0.796±0.005	0.895±0.018	0.711±0.023	0.857±0.002	0.821±0.013	0.808±0.003
GEM [15]	0.724±0.004	0.781±0.001	0.875±0.013	0.692±0.019	0.856±0.011	0.817±0.005	0.806±0.009
GROVER (base) [40]	0.700±0.001	0.743±0.001	0.812±0.030	0.664±0.032	0.826±0.007	0.673±0.018	0.625±0.009
GROVER (large) [40]	0.695±0.001	0.735±0.001	0.75±0.037	0.683±0.041	0.810±0.014	0.673±0.018	0.682±0.011
Molformer [56]	0.916±0.002	-	0.907±0.006	0.812±0.031	0.844±0.017	-	-
MolFormer-XL [41]	0.917±0.001	0.847±0.001	0.933±0.004	0.901±0.012	0.862±0.009	-	0.812±0.003
SubD-BERT (ours)	0.893±0.018	0.829±0.007	0.947±0.013	0.926±0.017	0.811±0.022	0.753±0.015	0.692±0.011
BERT+GIN (ours)	0.937±0.002	0.852±0.003	0.912±0.009	0.924±0.014	0.855±0.015	0.832±0.011	0.786±0.007
BERT+GCN (ours)	0.891±0.005	0.830±0.002	0.903±0.016	0.793±0.031	0.738±0.012	0.794±0.017	0.736±0.010
BERT+Graphormer (ours)	0.862±0.009	0.815±0.003	0.878±0.019	0.837±0.021	0.892±0.015	0.819±0.013	0.851±0.060
XGBoost (descriptors, ours)	0.821	0.663	0.856	0.871	0.695	0.650	0.562
LightGBM (descriptors, ours)	0.832	0.653	0.886	0.853	0.682	0.581	0.546
SVM (descriptors, ours)	0.612	0.617	0.525	0.679	0.547	0.559	0.534

MAE metric for the regression tasks

Models	Datasets					
	QM7	QM8 (E1-CC2)	QM9 (gap)	FreeSolv	ESOL	Lipo
MolCLR(GCN) [53]	85.4±2.7	0.0178±0.0003	0.0317±0.0005	3.259±0.261	1.419±0.040	0.957±0.010
MolCLR(GIN) [53]	91.6±3.1	0.0167±0.0004	0.0225±0.0003	2.884±0.249	1.253±0.037	0.651±0.004
ChemBERTa [6]	177.2±4.0	-	0.0317±0.0005	3.471±0.085	1.487±0.107	0.721±0.003
Uni-Mol [62]	41.8±0.2	0.0156±0.0001	0.0132±0.0003	1.480±0.048	0.788±0.024	0.603±0.010
GEM [15]	58.9±0.8	0.0171±0.0001	0.0246±0.0003	1.877±0.094	0.798±0.029	0.660±0.008
GROVER (base) [40]	94.5±3.8	0.0218±0.0004	0.0197±0.0003	2.186±0.052	0.983±0.090	0.817±0.008
GROVER (large) [40]	92.0±0.9	0.0224±0.0003	0.0186±0.0025	2.272±0.051	0.895±0.017	0.823±0.001
Molformer [56]	55.2±0.8	0.0095±0.0005	0.0139±0.0004	-	-	-
MolFormer-XL [41]	-	0.0102±0.0002	0.0164±0.0002	0.571±0.027	0.290±0.011	0.551±0.002
SubD-BERT (ours)	55.4±1.3	0.0126±0.0003	0.0148±0.0003	0.859±0.069	0.292±0.013	0.514±0.003
BERT+GIN (ours)	49.8±1.1	0.0083±0.0002	0.0145±0.0003	0.530±0.041	0.331±0.018	0.526±0.009
BERT+GCN (ours)	50.1±1.7	0.0098±0.0003	0.0166±0.0004	0.731±0.098	0.357±0.026	0.540±0.016
BERT+Graphormer (ours)	40.6±0.9	0.0114±0.0003	0.0134±0.0003	0.823±0.091	0.291±0.023	0.589±0.010
XGBoost (descriptors, ours)	69.2	0.0208	0.0174	5.040	1.233	1.013
LightGBM (descriptors, ours)	74.1	0.0203	0.0173	5.435	1.217	0.997
SVR (descriptors, ours)	143.4	0.0301	0.0334	6.207	1.830	1.171

- **Fragmentation:** Break molecules into chemically meaningful substructures (e.g., BRICS)
- **Descriptor Encoding:** Compute physicochemical descriptors for each fragment → form descriptor tokens
- **Text Encoder:** Process descriptor sequences using a Transformer (LM-style)
- **Graph Encoder:** Encode molecular graphs with a GNN (e.g., GIN or Graphormer)
- **Cross-Modal Alignment:** Use contrastive loss to align text and graph embeddings in a shared latent space
- **Unified Representation:** Combine both modalities for downstream molecular property prediction

- **CHEMIST** unifies graph and language views of molecules for richer representations
- Substructure-based descriptors bridge human chemical intuition and machine learning
- Cross-modal contrastive learning effectively aligns molecular structure and text
- Achieves **state-of-the-art performance** across classification and regression benchmarks
- Offers a general, interpretable framework for molecular representation learning

References

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